

→ Proof:

(1) follows from the ~~lemma~~ ^{prop} we prove about family of normal subgroups with trivial intersection.

(2) By the same Prop. as used in (1) we have that $\exists K \trianglelefteq G$ s.t. $\mathcal{G}(G/\text{Res}_\alpha(G)) = \mathcal{G}(G/K) \leq \alpha$. As the rank of $G/\text{Res}_\alpha(G)$ is a successor ordinal we get that $\mathcal{G}(G/\text{Res}_\alpha(G)) < \alpha$.

(3) $\text{Res}_2(G) \geq \text{Res}(G) \cap K$

Let $K \trianglelefteq G \Rightarrow G/K$ is compactly generated and discrete $\Rightarrow \mathcal{G}(G/K) \leq 1 \Rightarrow \text{Res}(G) \geq \text{Res}_2(G)$. ■

Proof: Let $\alpha \geq 2$ be an at most countable ordinal, G a t.d. l.c.s.c. group, and $(Q_i)_{i \in \mathbb{N}}$ be an exhaustion of G by compactly generated open subgroups. If α is a successor ordinal, then

$$\text{Res}_\alpha(G) = \bigcup_{i \in \mathbb{N}} \text{Res}_\alpha(Q_i).$$

If α is a limit ordinal, then

$$\text{Res}_\alpha(G) = \bigcap_{\beta < \alpha} \left(\bigcup_{i \in \mathbb{N}} \text{Res}_\beta(Q_i) \right).$$

→ Proof:

Take β an ordinal and $(Q_i)_{i \in \mathbb{N}}$ an $\subseteq$ -increasing sequence of G by compactly generated open subgroups. The union $\bigcup_{i \in \mathbb{N}} \text{Res}_\beta(Q_i)$ is an

increasing union of subgroups so $\bigcup_{i \in \mathbb{N}} \text{Res}_\beta(Q_i)$ is a subgroup of G .
 Given a compactly generated open subgroup L of G , sufficiently large i are s.t. $L \leq Q_i$, hence $\text{Res}_\beta(L) \leq \text{Res}_\beta(Q_i)$. We deduce that $\bigcup_{i \in \mathbb{N}} \text{Res}_\beta(Q_i)$ is in fact the union of $\text{Res}_\beta(L)$ as L ranges over all compactly generated open subgroups of G . Hence, $\bigcup_{i \in \mathbb{N}} \text{Res}_\beta(Q_i)$ is indeed a characteristic subgroup of G .

$\hookrightarrow$ every auto of G maps the subgroup to itself
 Suppose first $\alpha = \beta + 1$ is a successor ordinal and set $R := \bigcup_{i \in \mathbb{N}} \text{Res}_\alpha(Q_i)$. Suppose that $K \trianglelefteq G$ is closed with $\mathcal{G}(G/K) \leq \alpha$. By one of the last props $Q_i / (Q_i \cap K)$ is compactly generated open subgroup of G/K with rank at most α for every i . We deduce that $\text{Res}_\alpha(Q_i) \leq K$, so $R \leq K$. Therefore, $R \leq \text{Res}_\alpha(G)$.

For the converse inclusion, fix a compactly generated subgroup L/R of G/R . For i sufficiently, $L \leq Q_i R$, and there is a quotient map from $Q_i / \text{Res}_\alpha(Q_i)$ to $Q_i R / R$. Applying same prop as always

$$\mathcal{G}(L/R) \leq \mathcal{G}(Q_i R / R) \leq \mathcal{G}(Q_i / \text{Res}_\alpha(Q_i)) \leq \alpha$$

Thus, $\mathcal{G}(\text{Res}(L/R)) \leq \beta$, and since L is arbitrary, it follows that $\mathcal{G}(G/R) \leq \beta + 1 = \alpha$. We conclude that $R \geq \text{Res}_\alpha(G)$.

Suppose now α is a limit ordinal. Let $R_\beta := \text{Res}_\beta(G)$ (6). Since 6 has no quotients of rank exactly α ,

$$R_\alpha = \bigcap_{\beta < \alpha} R_\beta.$$

Induction on α and the first part of the prop applied to R_β imply

$$R_\alpha = \bigcap_{\beta < \alpha} \left(\overline{\bigcup_{i \in \mathbb{N}} \text{Res}_\beta(Q_i)} \right)$$

as required. ■

Coro:

Let 6 be a t.d.l.c. s.c. group and $(Q_i)_{i \in \mathbb{N}}$ be an $\ast$ increasing exhaustion of 6 by compactly open subgrps. Then

$$\text{Res}_2(G) = \overline{\bigcup_{i \in \mathbb{N}} \text{Res}_2(Q_i)}.$$

Obs: If α is a limit ordinal, it is not necessarily the case that $\text{Res}_\alpha(G) = \overline{\bigcup_{i \in \mathbb{N}} \text{Res}_\alpha(Q_i)}$. For instance, if 6 is a characteristically simple t.d.l.c. s.c. group with $\text{g}(G) = \omega + 1$ then $\text{Res}_\omega(G) = G$ but the finite rank $\ast$ radical of every compactly generated ^{open} subgroup of 6 is trivial.

Prop:

For G a non-compact elementary grp, one of the following occurs;

(1) $\mathcal{G}(G) = \mathcal{G}(\text{Res}_2(G)) = \alpha + 1$ where α is a countable limit ordinal;

(2) $\text{Res}_2(G)$ is a non-compact non-cocompact subgroup of G , and $\mathcal{G}(G) = \mathcal{G}(\text{Res}_2(G)) + 1$.

→ Proof:

Let $(Q_i)_{i \in \mathbb{N}}$ be an increasing exhaustion of G by compactly generated open subgroups and set $R := \text{Res}_2(G)$. We have by the last coroll. $\bigcup_{i \in \mathbb{N}} \text{Res}(Q_i) = R$. We also have

$$\mathcal{G}(G) = \sup_{i \in \mathbb{N}} \mathcal{G}(\text{Res}(Q_i)) + 1,$$

and by some prop we have

$$\mathcal{G}(R) = \sup_{i \in \mathbb{N}}^+ \mathcal{G}(\text{Res}(Q_i)).$$

Set $\beta := \sup_{i \in \mathbb{N}} \mathcal{G}(\text{Res}(Q_i))$. If β is a successor ordinal, then $\mathcal{G}(R) = \beta$ and $\mathcal{G}(G) = \beta + 1$, so $\mathcal{G}(G) = \mathcal{G}(R) + 1$. In particular, R is a proper subgroup of G with $\mathcal{G}(G/R) = 2$. Because cocompact normal subgroups have same rank as the group, R can be cocompact if it is trivial; however this would imply that G is compact, contradicting the hypothesis. We conclude (2) holds but (1) does not hold.

If β is a limit ordinal $\Rightarrow \beta$ is Transfinite. We conclude $\mathcal{G}(\beta) = \mathcal{G}(\beta) = \beta + 1$. Condition (1) holds and (2) does not hold, completing the proof of the dichotomy. ■

We now prove the theorem about characteristically simple t.d. l.c.s.c. groups

→ Proof:

Suppose (3) does not hold $\Rightarrow G$ is elementary. Since G is characteristically simple, $\text{Res}_2(G) = \{1\}$, or $\text{Res}_2(G) = G$. By ~~last prop~~ last prop $\text{Res}_2(G) = \{1\}$ ~~iff~~ iff G is elementary of rank 2; the case $\mathcal{G}(G) = 1$ being eliminated by the hypothesis that $G \neq \{1\}$. We thus deduce that (1) holds if $\text{Res}_2(G) = \{1\}$. If $\text{Res}_2(G) = G$, then G is non-compact, since $\mathcal{G}(G) > 2$. The dichotomy of last prop thus ensures that $\mathcal{G}(G) = \alpha + 1$ for α some transfinite limit ordinal, hence (2) holds. ■